

AI for Medicine – Project Report

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1. Project Title

A Multimodal Fusion Network for Keratoconus Detection: Combining Orbscan Corneal Maps and Clinical Biomarkers

2. Problem Statement

Keratoconus (KC) is a progressive, non-inflammatory disorder characterized by corneal thinning and a conical corneal shape. These morphological changes result in irregular astigmatism, myopia, and visual impairment, and may ultimately necessitate corneal transplantation (1). The onset of KC typically occurs during adolescence or early adulthood. Early detection is critical, as corneal cross-linking (CXL)—the only intervention capable of halting disease progression—is effective only prior to significant structural damage (2). Accurate diagnosis requires integrating multiple corneal topography maps with clinical biomarkers, including Kmax, pachymetry, and asphericity. This diagnostic process is time-consuming and subject to inter-observer variability (3) (4). AI-based approaches most rely on a single modality in isolation, failing to exploit the complementary diagnostic information that clinicians integrate in practice, (5) motivating the development of a multimodal fusion approach that jointly processes corneal images and structured clinical features.

3. Objective of the Study

This study aims to develop and evaluate a multimodal deep learning framework for binary classification of keratoconus using the CornOrb dataset (4). The proposed MultimodalFusionNet model combines four Orbscan corneal topography maps with seven clinical biomarkers using a dual-branch fusion architecture. A secondary objective is to quantify each modality's contribution via ablation, comparing the fused model against image-only and clinical-only baselines.

4. Dataset Description

The CornOrb dataset (4) is a publicly available multimodal resource released on Zenodo (6), comprising 1,454 eyes from 744 patients collected retrospectively at a single clinical center in Tlemcen, Algeria using the Orbscan 3 corneal topographer. For each eye, four corneal topography maps (axial curvature, anterior elevation, posterior elevation, and pachymetry) are provided alongside structured clinical annotations including age, gender, astigmatism, maximum keratometry (Kmax), central and thinnest pachymetry, and anterior and posterior asphericity. Labels are binary, distinguishing normal corneas (n=889) from keratoconus cases (n=565), reflecting a moderate class imbalance of approximately 61/39.

5. Data Preprocessing

The CornOrb dataset (4) required minimal preprocessing owing to its clean, standardized release format. No missing values were present in the clinical annotations, and all corneal images were provided as fixed-resolution PNG files.

For the image branch, each of the four corneal maps was resized to 224×224 pixels and normalized using ImageNet (7) mean and standard deviation (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), consistent with transfer learning from pretrained ResNet (8) weights.

Data augmentation was applied exclusively to the training set. It included random horizontal flipping, rotation, and color jitter to improve generalization.

For the clinical branch, the seven structured features (age, astigmatism, Kmax, central pachymetry, thinnest pachymetry, anterior asphericity, and posterior asphericity) were standardized to zero mean and unit variance using a StandardScaler fitted solely on the training set and applied to the validation and test sets

6. Avoiding Data Leakage

We took several steps to prevent data leakage.

First, we used the dataset's predefined 80/20 split to create the test set, which was kept separate from the training set during model development and hyperparameter tuning.

Second, we ensured that both eyes of the same patient were assigned exclusively to a single partition, preventing inter-eye correlation from leaking across splits; a critical consideration given that keratoconus is a bilateral disease and fellow eyes of the same patient exhibit strongly correlated corneal morphology in both pachymetry and posterior elevation (9). We checked that there was no patient overlap between splits by confirming that the intersection of patient identifiers across the train, validation, and test sets was empty.

Finally, as it already mentioned, we fitted the StandardScaler for clinical feature normalization only on the training data, then applied it to the validation and test sets.

7. Machine Learning Pipeline

7.1 Model Architecture

The proposed MultimodalFusionNet is a dual-branch deep learning architecture that jointly processes corneal topography images and structured clinical features.

The image branch uses a pretrained ResNet-18 convolutional neural network as a feature extractor, processing four corneal maps concatenated along the channel dimension (yielding a 12-channel input) and projecting the extracted representations into a fixed-size embedding vector.

The clinical branch comprises a lightweight fully connected network with batch normalization and dropout regularization that processes seven standardized clinical biomarkers into a corresponding embedding space.

The embeddings from both branches are concatenated and passed through a final classification head to generate binary predictions (Normal versus Keratoconus).

7.2 Training Configuration

The model was trained using the Adam optimizer with a learning rate scheduler and cross-entropy loss. Dropout regularization was implemented in both branches to mitigate overfitting. Data augmentation, including random horizontal flipping, rotation, and color jitter, was applied exclusively to the training set images to enhance generalization.

7.3 Validation Strategy

The dataset was divided using a predefined 80/20 patient-level split for evaluation. Additionally, 15% of the training patients were held out as a validation set for early stopping and model selection.

K-fold cross-validation was not performed at test time, as the predefined split was intended to ensure reproducibility and enable fair benchmarking against future work on CornOrb.

7.4 Ablation Study

An ablation study was conducted to isolate the contribution of each modality by training and evaluating three model variants under identical conditions: image-only, clinical-only, and fused.

7.5 Evaluation Metrics

Performance was assessed using ROC-AUC as the primary metric, along with accuracy, precision, recall, and F1-score for each class.

Keratoconus recall was prioritized as the most clinically significant metric, since false negatives, or missed keratoconus cases, have greater clinical consequences than false positives in a screening context.

8. Results

8.1 Training Dynamics

All three model variants converged stably over 30 epochs. Final validation accuracies were comparable across all three variants (~97%). Training curves for validation loss and accuracy across all three model variants are shown in Figure 1.

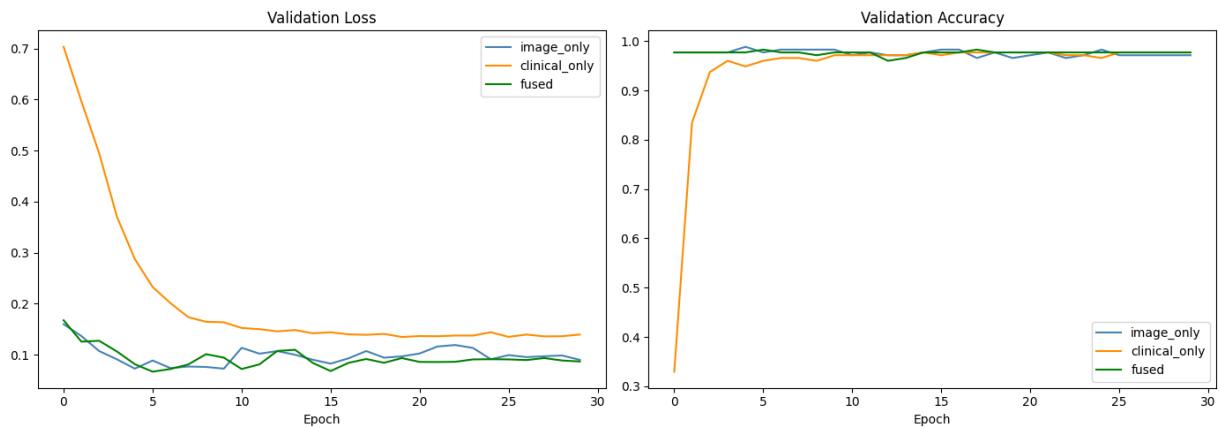


Figure 1

8.2 Test Set Performance

Table 1 summarizes the classification performance of all three model variants on the held-out test set (n=291; 178 Normal, 113 Keratoconus).

Model	ROC-AUC	Accuracy	Precision (KC)	Recall (KC)	F1 (KC)
Image-Only	0.996	0.97	0.97	0.94	0.95
Clinical-Only	0.996	0.95	1.00	0.86	0.92
Fused	0.999	0.98	0.99	0.95	0.97

Table 1

8.3 ROC Curves and Confusion Matrices

ROC curves for all three model variants are shown in Figure 2.

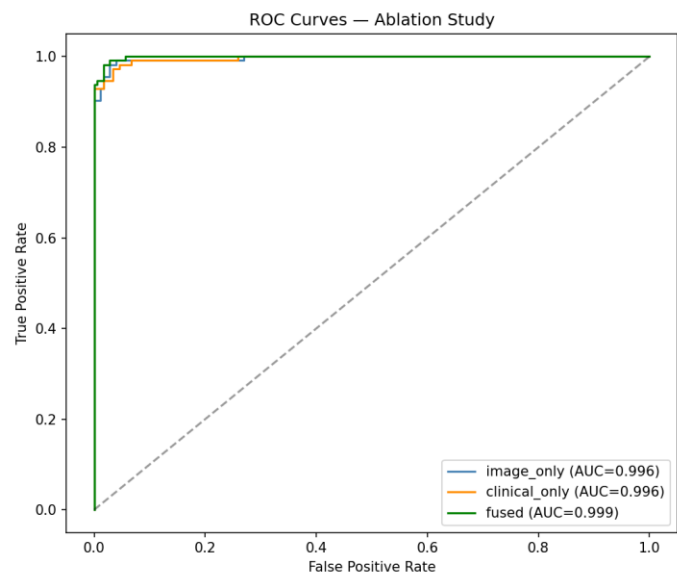


Figure 2

Confusion matrices are shown in Figure 3.

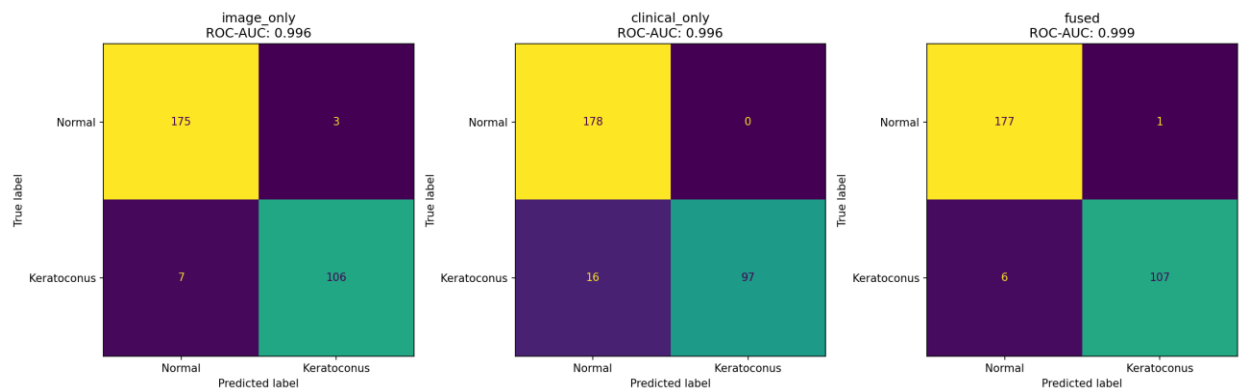


Figure 3

8.4 Misclassified Cases

Of the 7 misclassified cases produced by the fused model, 6 were false negatives (missed keratoconus) and 1 was a false positive. A comparison of misclassified clinical feature profiles against the correctly predicted ones are summarized in Table 2.

Feature	Misclassified Mean	Misclassified Std	Correct Mean	Correct Std	Mean Diff
Age (years)	0.386	1.498	-0.016	0.979	+0.402
Astigmatism (D)	-0.824	1.175	0.068	1.232	-0.891
Kmax (D)	0.138	0.387	0.041	1.029	+0.096
Central Pachymetry (μm)	0.225	1.016	0.197	0.974	+0.028
Thinnest Pachymetry (μm)	0.086	1.187	0.174	1.038	-0.088
Anterior Asphericity	0.515	0.784	0.008	1.047	+0.507
Posterior Asphericity	0.238	0.496	-0.042	0.926	+0.280

Table 2

Note: all values are z-scored (standardized), so means and standard deviations are relative to the training set distribution, not raw clinical units.

8.5 Grad-CAM Visualizations

Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to selected cases from the fused model's test set predictions. Four sets of visualizations are presented.

Grad-CAM maps for the top 4 most confidently misclassified keratoconus eyes:

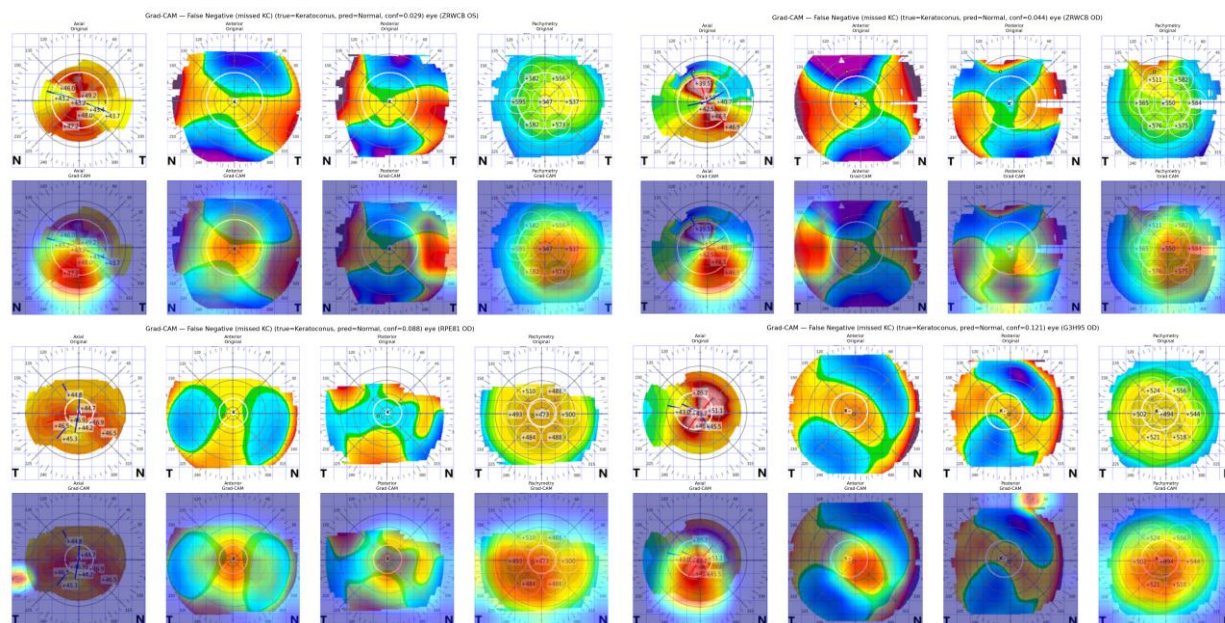


Figure 4

Grad-CAM maps for the top 4 correctly classified keratoconus eyes:

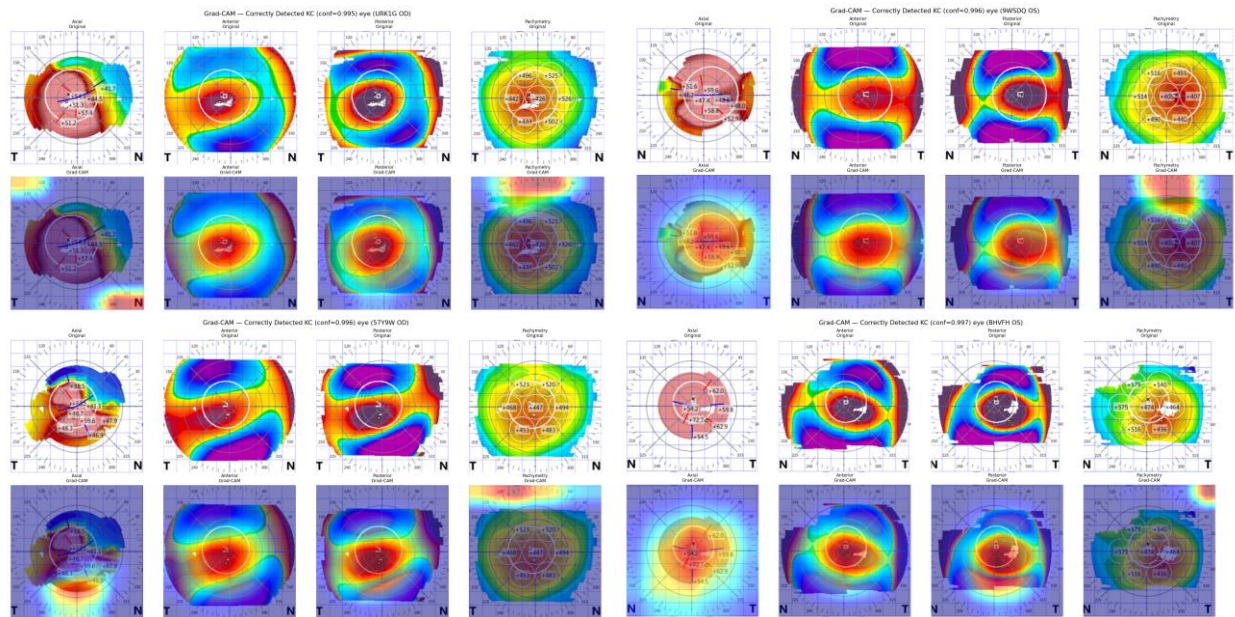


Figure 5

Grad-CAM map for the single false positive case:

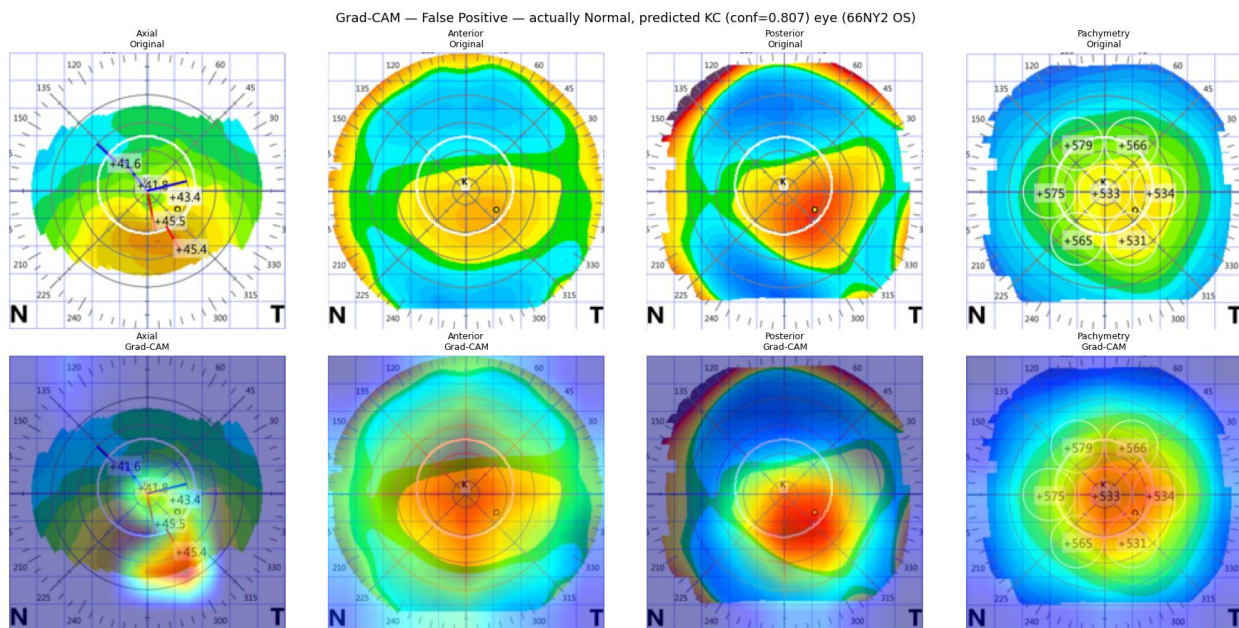


Figure 6

Grad-CAM maps for the top 4 correctly classified normal eyes:

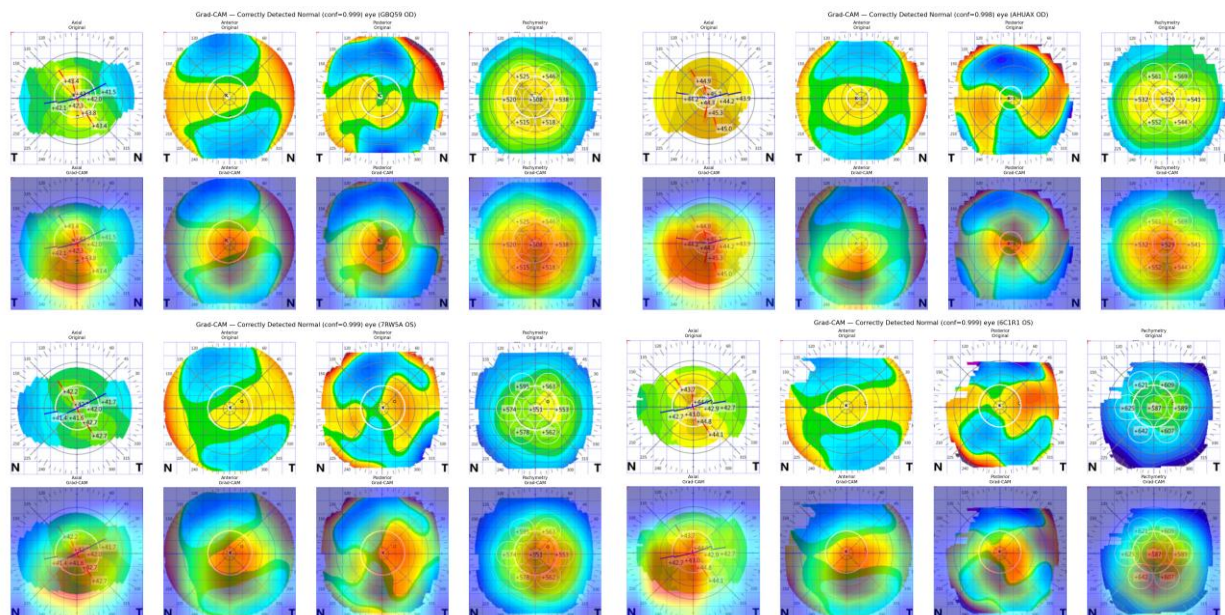


Figure 7

9. Discussion

9.1 Interpretation of Results

The fused model outperformed both unimodal baselines, achieving a ROC-AUC of 0.999 and 98% accuracy with only 7 misclassifications out of 291 test cases. The contrasting error profiles of the unimodal models explain why fusion helps: the clinical-only model never falsely flagged a normal eye but missed 16 keratoconus cases (recall: 0.86), while the image-only model distributed errors more evenly. The fused model inherited the strengths of both, reducing false negatives to 6 — the most critical error type in a screening context.

The 6 missed keratoconus cases share a clinically coherent profile: near-normal Kmax, preserved pachymetry, and elevated anterior asphericity, consistent with early or subclinical disease where discriminative features are not yet pronounced. The single false positive exhibited atypically high astigmatism and asphericity values, resembling a keratoconus-like corneal shape despite a normal label. Notably, 5 out of 7 misclassified cases were right eyes (OD), though given the small error count this is unlikely to reflect a systematic laterality bias and warrants no strong conclusion.

9.2 Grad-CAM Analysis

When the model accurately detected keratoconus, it focused exactly where eye doctors do: the center of the cornea. It successfully spotted classic disease signs, like the steep cone shape and the thinnest parts of the eye (down to 376–436 μm). This proves the AI learned real medical features rather than random image patterns.

The model missed a few cases because the disease was too mild or in its very early stages. In these eyes, the thickness and shape looked mostly normal without any obvious problem areas to trigger a warning. For example, patient ZRWCB was missed in both eyes due to these highly subtle symptoms.

The model mistakenly flagged one healthy eye as diseased (Patient 66NY2). A tricky combination of mild shape irregularities and a slightly thin cornea (533 μm) mimicked early keratoconus. This highlights how difficult it can be for the AI to tell a slightly unusual normal eye from an early-stage diseased eye.

For healthy eyes, the model correctly recognized broad, even, and regular patterns. These corneas had normal shape and healthy thickness (508–551 μm), lacking any of the concentrated "hot spots" that indicate structural damage.

9.3 Limitations

- **Single-center, single-device data:** All images were acquired with the Orbscan 3 at one Algerian clinic, limiting generalizability to other devices (e.g., Pentacam) or populations.
- **Cross-sectional design:** The model cannot distinguish stable from progressive keratoconus, which is clinically important for treatment decisions.
- **Dataset size:** With 744 patients, performance estimates may vary on larger or more diverse cohorts. External validation remains necessary.
- **Class imbalance:** The 61/39 split was handled through stratified splitting only; explicit reweighting strategies were not explored.
- **Subclinical keratoconus:** The model's primary failure mode is early disease where topographic and clinical features have not yet reached the severity thresholds learned during training, an inherent limitation shared by most current AI-based keratoconus detection systems.

10. Conclusions and Future Work

This study presents **MultimodalFusionNet**, a dual-branch deep learning framework for keratoconus classification that integrates four Orbscan topography maps with seven clinical biomarkers. Evaluated on the CornOrb dataset (N=291), the model achieved 98% accuracy and a ROC-AUC of 0.999, outperforming both image-only and clinical-only baselines.

Ablation studies demonstrated that the modalities are highly complementary: clinical data drove high precision while imaging enhanced sensitivity, reducing total misclassifications to just seven cases. Furthermore, Grad-CAM analysis confirmed the model's clinical interpretability by accurately highlighting the cone apex and thinnest corneal zones. Consistent with challenges across the broader keratoconus AI literature, the model's primary limitation remains the detection of subclinical or early-stage disease where focal topographic deformation is absent.

Future Work

To address current limitations and enhance real-world clinical deployment, future research should focus on:

- **Robustness and Generalization:** Conducting external validation on independent, multi-device datasets (e.g., Pentacam cohorts) to ensure the model generalizes across different populations and topographers.
- **Early-Stage Detection:** Improving sensitivity to subclinical cases by integrating additional parameters, such as corneal biomechanical indices, and exploring techniques like contrastive learning or focal loss to handle class imbalance.
- **Expanded Clinical Utility:** Transitioning from cross-sectional binary classification to longitudinal modeling to track disease progression over time. Additionally, expanding the model to perform multi-class severity grading (e.g., suspect, mild, moderate, severe) would provide more actionable insights for clinical decision-making.

11. Ethics and Data Privacy

The CornOrb dataset is publicly available under a Creative Commons Attribution 4.0 International (CC BY 4.0) license (10) via Zenodo (6), permitting academic and research use with attribution. All data were fully anonymized prior to release by the original authors, requiring no additional ethics approval for this project. All libraries used (PyTorch, scikit-learn, torchvision) are open-source and permissively licensed. No model outputs were used to inform any real clinical diagnosis; this work was conducted solely for academic purposes.

12. Code and Reproducibility

All experiments were implemented in Python using Google Colab and are available in the notebook CornOrb_Multimodal_Keratoconus.ipynb, hosted at [Github](#).

The repository includes the trained model weights (best_image_only.pt, best_clinical_only.pt, best_fused.pt), allowing results to be reproduced without retraining by setting RUN_MODE = "eval_only" in the notebook.

The CornOrb dataset can also be downloaded directly from its original source at Zenodo (6). Alternatively, if the dataset is already available on Google Drive, it can be loaded by setting DATA_SOURCE = "drive" in the notebook.

All required libraries (torch, torchvision, scikit-learn, pandas, numpy, matplotlib, seaborn, Pillow) come pre-installed in Google Colab; the only additional dependency is grad-cam, which is installed automatically at the start of the notebook via. All experiments were run on a Google Colab T4 GPU.

13. References

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